

MACIEJ STEMPIAK

SOFTWARE ENGINEER / C++20 & .NET

Backend and systems engineer with production experience in 5G RAN software and backend services for Poland's national e-invoicing platform. Focused on reliable, testable and performance-sensitive systems.



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EXPERIENCE

Backend Developer (.NET)

May 2026 - present

Symfonia | KSeF platform

- Build and debug C#/.NET and GraphQL services using SQL, Hangfire, Dapr, Kubernetes and OPA.
- Contributed to the KSeF API 1.0 -> 2.0 migration across endpoints, data models, authentication, persistence, GraphQL schemas and Ministry of Finance integration.
- Investigate production failures and create automated tests, test harnesses, repository documentation and validation workflows.

C++ Software Engineer

Jul 2025 - Apr 2026

Nokia | 5G RAN | Internship followed by extended contract

- Developed C++20 decision logic for temperature-aware cell activation, traffic control and energy management in 5G base stations.
- Extended operator-configuration handling for Massive MIMO and beamforming functionality.
- Added automated tests, validated changes in simulation, and debugged behavior using GDB and Valgrind.

TECHNICAL STRENGTHS

Production

C++20, C#/.NET, GraphQL, SQL, Dapr, Kubernetes, OPA

Tooling

Python, WebAssembly, Linux, GDB, Valgrind, GoogleTest, xUnit, CI

EDUCATION

BSc in Computer Science

2022 - 2026

University of Wrocław

- Thesis: implementation of the Catan board game and AI agents for one-versus-one gameplay. Defended in March 2026.
- Selected coursework: Concurrent Programming in C++, Algorithms and Data Structures, Computer Architecture.

SELECTED ENGINEERING PROJECTS

Two-Player Catan Simulation Engine & Game AI

GitHub

Co-authored BSc thesis: a bit-packed C++20 simulation engine and reproducible tournament platform for 13 native agents.

Results: 59% lower per-game cost; 250-300 games/s; board state under 800 B; 242 generated test cases; GCC, MSVC, Clang and AddressSanitizer CI.

Bot Arena

GitHub

Python tournament framework with isolated bot processes, execution time limits, structured match logs and move-by-move replay.

SpaceX Launch Data & Recommendation API

GitHub

.NET 10 Minimal API with idempotent imports, EF Core/SQLite persistence, bounded parameters, upstream timeout handling and automated tests.

Applied ML research

YOLOv8 and PyTorch experiments for melanoma detection in histopathology imagery and glaucoma detection in retinal fundus data.

ACHIEVEMENTS

- Rector's Scholarship for Academic Achievement, 2023/24. GPA 4.74/5.00; ranked 11th of 160 in the first-cycle Computer Science scholarship ranking.
- Honorable Mention, first stage of the 29th Polish Olympiad in Informatics.

TEACHING AND LEADERSHIP

- Mathematics and Computer Science Tutor, since 2020. Worked one-to-one with more than 20 students, including Matura preparation.
- Certified Sailing Instructor, 2023-2025. Taught adults and young people over three summer seasons and prepared participants for the Polish yacht-sailing licence examination.

LANGUAGES

PolishNative

EnglishCambridge C1 Advanced | Grade B | 198/210

ADDITIONAL TRAINING

10xDevs 3.0

2026

Practical programme covering agent-assisted software development, testing and workflow automation.